Eigenvalue multiplicity of the wreath product Cayley graphs

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Cayley graphs

Let G be a finite group and let $\mathcal{S} = \mathcal{S}^{-1} \subseteq G \setminus \{e\}$.

$$\Gamma = \mathrm{Cay}(G, \mathcal{S}): \quad V(\Gamma) = G, \quad g \sim gs \ (s \in \mathcal{S}).$$

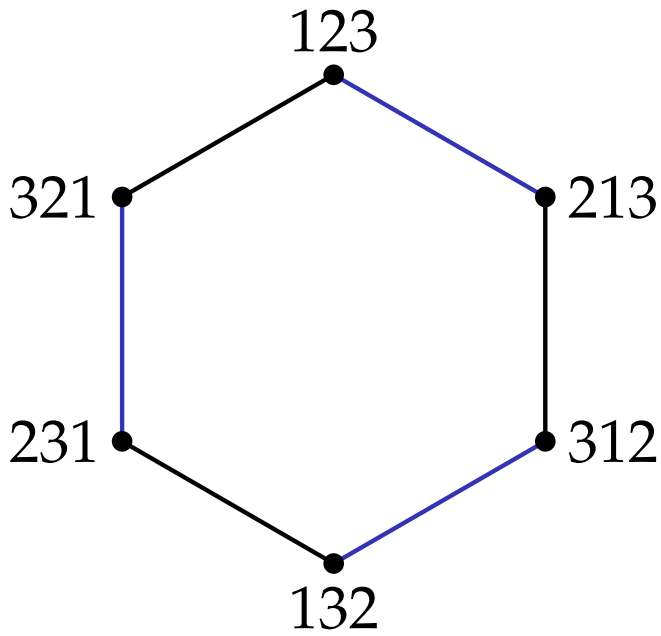
Example: $G = S_3$

$$\mathcal{S} = \{(12), (13)\}.$$

Vertices: the six permutations.

(12): swap entries 1 and 2.

(13): reverse all three entries.



Pancake graphs

Take $G = S_n$. The prefix reversal r_k reverses the first k positions, for $2 \leqslant k \leqslant n$. For example,

$$(3,1,4,2) \xrightarrow{r_3} (4,1,3,2).$$

Each flip is its own inverse: $r_k^2 = e$.

These generators define the pancake graph

$$P_n = \mathrm{Cay}(S_n, \{r_2, \ldots, r_n\}).$$

It has $n!$ vertices and $n-1$ neighbours at each vertex.

For $n = 4$: 24 vertices and three flips, r_2, r_3, r_4.

From flips to integer eigenvalues

Dalfó–Fiol (2020) obtain integer eigenvalues of P_n by summing the permutation matrices of the flips.

Each flip also permutes coordinates. For $n = 4$:

A single flip	Flip matrix	Sum
	$R(r_3)$	$M_4 = \sum_{k=2}^{4} R(r_k)$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \xrightarrow{r_3} \begin{pmatrix} x_3 \\ x_2 \\ x_1 \\ x_4 \end{pmatrix} \quad = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}$$

$$\mathrm{Spec}(M_4) = \{-1, 0, 2, 3\} \subseteq \mathrm{Spec}(P_4).$$

For general n, write $M_n = \sum_{k=2}^{n} R(r_k)$.

The standard S_n representation

For $n = 4$, compare two eigenvectors of M_4:

Constant vector $\mathbf{1}$

$$\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \xrightarrow{M_4} \begin{pmatrix}3\\3\\3\\3\end{pmatrix}$$

$\lambda = 3$

Sum-zero vector v

$$\begin{pmatrix}0\\1\\-1\\0\end{pmatrix} \xrightarrow{M_4} \begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

$\lambda = 0$

$$\mathbb{C}^n = \mathbb{C}\mathbf{1} \oplus W_n, \qquad W_n = \{v : \textstyle\sum_i v_i = 0\}.$$

The action on W_n is the standard S_n representation.

Dalfó–Fiol (2020) obtain, for $n \geqslant 2$:

$$\mathrm{Spec}(M_n|_{W_n}) = \{-1, 0, \ldots, n-2\} \setminus \{\lfloor n/2 \rfloor - 1\}.$$

The full M_n is a quotient matrix: cells fix the position of 1.

Signed and coloured permutations

A signed flip ($m = 2$)

$$(-3, 1, 2) \xrightarrow{\text{reverse first two}} (1, -3, 2) \xrightarrow{\text{change signs}} (-1, 3, 2).$$

A coloured flip ($m = 3$): superscripts record colours.

$$(3^1, 1^0, 2^0) \xrightarrow{\text{reverse first two}} (1^0, 3^1, 2^0) \xrightarrow{\text{add } 1} (1^1, 3^2, 2^0).$$

In general, $r_k^{\pm}$ reverses the first k entries and adds ± 1 to their colours modulo m.

These permutations form the wreath product $\mathbb{Z}_m \wr S_n$.

$$\mathcal{P}_m(n) = \mathrm{Cay}\big(\mathbb{Z}_m \wr S_n, \{r_k^{\pm} : 1 \leqslant k \leqslant n\}\big) \quad (m \geqslant 2).$$

$m = 2$: the burnt pancake graph on B_n.

Groups behind wreath product Cayley graphs

The generalised symmetric group is

$$S(m,n) = \mathbb{Z}_m \wr S_n, \qquad B_n = S(2,n).$$

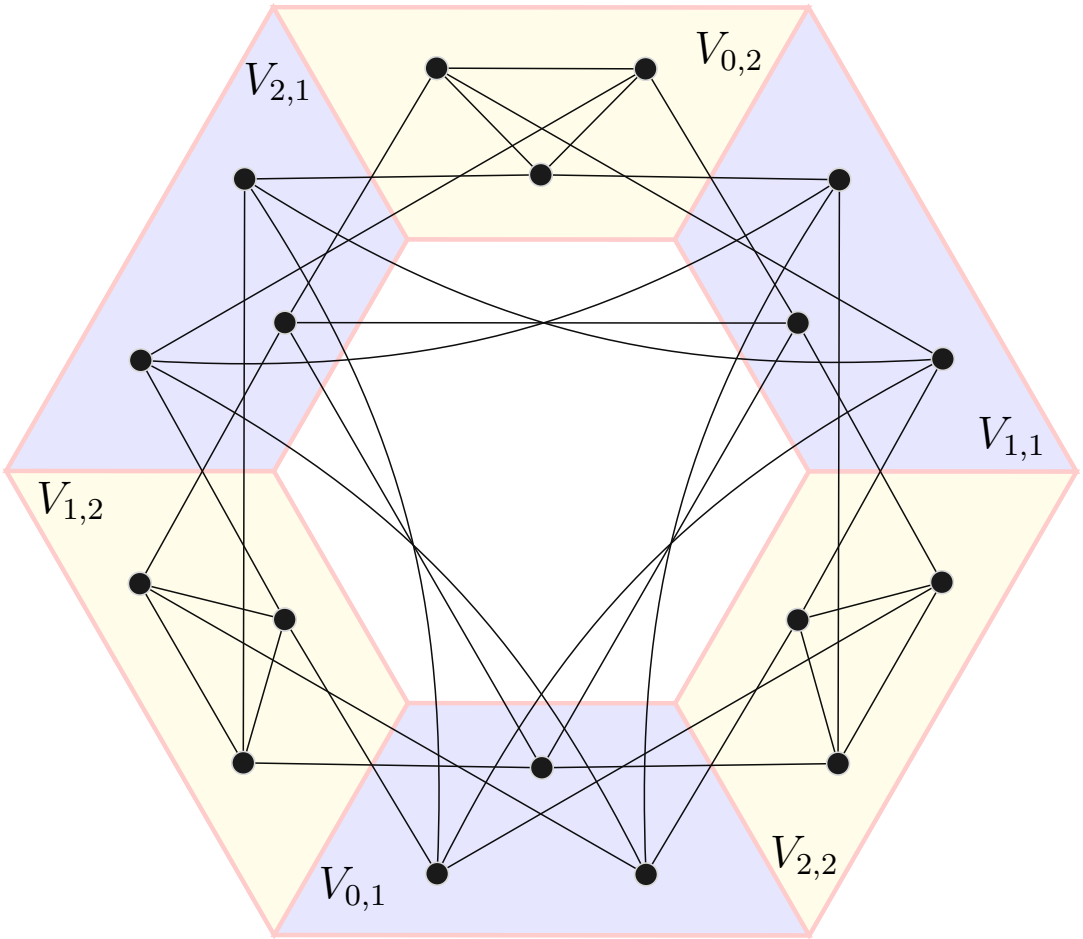
Group	Letters	Permutation matrix
S_n	n ordinary letters	$n \times n$
B_n	$2n$ signed letters	$2n \times 2n$
$S(m,n)$	mn coloured letters	$mn \times mn$

Forgetting colours recovers the ordinary permutation:

$$(3^1, 1^0, 2^0) \longmapsto (3,1,2) \in S_3.$$

The underlying S_n action on the sum-zero space W_n is the standard representation.

A generalised pancake graph



$\mathcal{P}_3(2)$

18 vertices
6 cells

An equitable partition

Each cell fixes the colour and position of the letter 1.

Integer eigenvalues and multiplicities

Blanco and Buehrle (2024, 2025) use Dalfó–Fiol's method with the actions on signed and coloured letters.

Write $T_n = M_n + I$, including the ordinary identity flip $r_1 = e$.

$$\text{Burnt (2024):} \quad \operatorname{Spec}(T_n) \subseteq \operatorname{Spec}(\mathcal{P}_2(n)),$$

$$\text{Coloured (2025):} \quad 2\operatorname{Spec}(T_n) \subseteq \operatorname{Spec}(\mathcal{P}_m(n)) \quad (m \geqslant 3).$$

Conjecture (Blanco and Buehrle, 2025)

They also conjecture lower bounds on the multiplicities of these integer eigenvalues.

Order of the vertices

For $\mathcal{P}_3(2)$, label each quotient vertex by (c, j):
c is the colour and j the position of the letter 1.

Position first

$(0,1),(1,1),(2,1),(0,2),(1,2),(2,2)$

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ \hline 0 & 1 & 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 & 0 & 2 \end{array}\right)$$

Colour first

$(0,1),(0,2),(1,1),(1,2),(2,1),(2,2)$

$$\left(\begin{array}{cc|cc|cc} 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ \hline 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 & 0 \\ \hline 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 2 \end{array}\right)$$

Ordering by colour reveals the block-circulant form.

The multiplicity theorem

Theorem (Greaves and Zhu, 2026)

Let $4 \mid m$ and $n \geqslant 2$. Then, for $1 \leqslant k \leqslant n-1$,

$$\operatorname{mult}_{\mathcal{P}_m(n)}(2k) \geqslant \begin{cases} 3, & k \neq \lfloor n/2 \rfloor, \\ 2, & k = \lfloor n/2 \rfloor. \end{cases}$$

The block-circulant quotient matrix $Q_{m,n}$ satisfies

$$Q_{m,n} \sim 2T_n \oplus 2D_n \oplus 2D_n \oplus \cdots,$$

where $D_n = \operatorname{diag}(0, 1, \ldots, n-1)$.

Two $2D_n$ blocks — Two copies of each $2k$

Standard S_n contribution — A third, except at $2\lfloor n/2 \rfloor$

Representation-theoretic interpretation

Complex irreducible representations are indexed by:

Group	Labels of total size n
S_n	Partitions λ
B_n	Bipartitions (λ, μ)
$S(m,n)$	Coloured partitions $(\lambda^{(0)}, \ldots, \lambda^{(m-1)})$

An S_n representation extends by forgetting colours:

$$\lambda \longmapsto (\lambda, \varnothing, \ldots, \varnothing), \qquad \text{with the same dimension.}$$

The first component corresponds to the trivial colour action.

Standard representation: $\lambda = (n-1, 1)$, dimension $n-1$.

Its flip-sum eigenvalues occur in the coloured Cayley graphs.

Thank you for your attention

Further reading:

G. Greaves and H. Zhu, *A note on some spectral properties of generalised pancake graphs*, Discrete Math. **349**(9) (2026), 115102.